Grazioso Salvare README

About the Project/Project Title

This is a full-stack web application dashboard created for Grazioso Salvare, a search and rescue dog training organization. The dashboard uses a python CRUD module to connect with a mongo database allowing users to filter the Austin Animal Center shelter data in Austin, Texas. The dashboard provides a user-friendly way to identify dogs that meet certain profiles based on their age, sex and so on. The projects architecture uses the MVC design pattern, MongoDB serves as the Model, Dash widgets as the View, and the Python CRUD module as the Controller

Getting Started

You will need the following software or libraries:

Python

Python CRUD module(add link to readME)

PyMongo

A Mongo database

Dash

Dash_leaflet

Pandas

Required Functionality

Here we can see the required functionality with various screenshots:

No filter

Grazioso Salvare Dashboard
Created by: Creston Getz

None

Select a button on the left to update the geolocation map:

REC_NUM	AGE_UPON_OUTCOME	ANIMAL_ID	ANIMAL_TYPE	BREED	COLOR	DATE_OF_BIRTH	DATETIME	MONTHYEAR	NAME	OUTCOME_SUBTYPE	OUTCOME_TYPE	SEX_UPON_OUTCOME	LOCATION_LAT	LOCATION_LONG
Search														
2	1 year	A725717	Cat	Domestic Shorthair Mix	Silver Tabby	2015-05-02	2016-05-06 10:49:00	2016-05-06T10:49:00		SCRP	Transfer	Spayed Female	30.6525984560228	-97.7419963476444
3	2 years	A716330	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-18	2015-12-28 18:43:00	2015-12-28T18:43:00	Frank		Adoption	Neutered Male	30.7595748121648	-97.5523753807133
1	3 years	A746874	Cat	Domestic Shorthair Mix	Black/White	2014-04-10	2017-04-11 09:00:00	2017-04-11T09:00:00		SCRP	Transfer	Neutered Male	30.5066578739455	-97.3408780722188
5	2 years	A691584	Dog	Labrador Retriever Mix	Tan/White	2012-11-06	2015-05-30 13:48:00	2015-05-30T13:48:00	Luke		Return to Owner	Neutered Male	30.7104815618433	-97.562297435286
4	7 months	A733653	Cat	Siamese Mix	Seal Point	2016-01-25	2016-08-27 18:11:00	2016-08-27T18:11:00	Kitty		Adoption	Intact Female	30.3188063374257	-97.7240376703891
6	5 years	A696004	Dog	Cardigan Welsh Corgi Mix	Sable/White	2010-01-27	2015-01-28 10:39:00	2015-01-28T10:39:00	Lucy	Rabies Risk	Euthanasia	Spayed Female	30.6737365854231	-97.707971529467
9	3 years	A720214	Dog	Labrador Retriever Mix	Red/White	2013-02-04	2016-02-11 12:41:00	2016-02-11T12:41:00	Blessing		Adoption	Spayed Female	30.3870648199411	-97.3684339731375
8	1 year	A736551	Dog	Labrador Retriever/Australian Cattle Dog	Black	2015-10-12	2016-11-27 18:00:00	2016-11-27T18:00:00	*Mia		Adoption	Spayed Female	30.4443212820182	-97.7326980338793
10	3 months	A664290	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08T14:58:00	*Taylor		Adoption	Spayed Female	30.7583105481048	-97.618292198845
11	1 year	A721199	Dog	Dachshund Wirehair Mix	Tan/White	2015-02-23	2016-02-27 17:49:00	2016-02-27T17:49:00	Belle		Adoption	Spayed Female	30.7290272761146	-97.3753328216134

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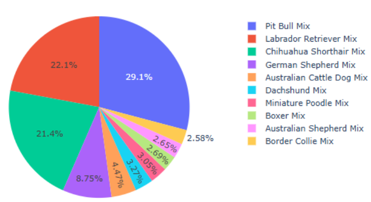
Top 10 Dog Breed Distribution



Water Rescue Filter

Grazioso Salvare Dashboard
Created by: Creston Getz

Water Rescue

Select a button on the left to update the geolocation map:

REC_NUM	AGE_UPON_OUTCOME	ANIMAL_ID	ANIMAL_TYPE	BREED	COLOR	DATE_OF_BIRTH	DATETIME	MONTHYEAR	NAME	OUTCOME_SUBTYPE	OUTCOME_TYPE	SEX_UPON_OUTCOME	LOCATION_LAT	LOCATION_LONG	AGE_UPON_OUTCOM
Search															
36	6 months	A706953	Dog	Labrador Retriever Mix	Yellow	2014-12-06	2015-07-06 11:33:00	2015-07-06T11:33:00		Medical	Euthanasia	Intact Female	30.5480802368633	-97.2969969058957	30.3544642857143
732	2 years	A749782	Dog	Labrador Retriever Mix	Tan/White	2015-05-19	2017-07-25 14:59:00	2017-07-25T14:59:00	*Catalina		Return to Owner	Intact Female	30.6138310636757	-97.5752164857665	114.089186507937
1121	1 year	A757158	Dog	Labrador Retriever Mix	White/Black	2016-08-30	2017-08-31 14:12:00	2017-08-31T14:12:00	Pirata		Return to Owner	Intact Female	30.5572161697962	-97.5363224263878	52.3702380952381
1628	9 months	A740471	Dog	Labrador Retriever Mix	Tan/White	2016-03-17	2016-12-23 17:13:00	2016-12-23T17:13:00	Mika		Adoption	Intact Female	30.7569243032341	-97.7392549176654	40.2453373015873
1757	7 months	A742767	Dog	Labrador Retriever Mix	Black	2016-06-27	2017-02-14 15:20:00	2017-02-14T15:20:00	Marley		Return to Owner	Intact Female	30.4869754937324	-97.4280017197358	33.234126984127
1988	1 year	A762781	Dog	Labrador Retriever Mix	Black/White	2016-11-27	2017-12-03 13:09:00	2017-12-03T13:09:00		Partner	Transfer	Intact Female	30.2840111162863	-97.4600542219677	53.0782738095238
2041	2 years	A702745	Dog	Labrador Retriever Mix	Black	2013-05-22	2015-05-22 11:45:00	2015-05-22T11:45:00	Abigail		Return to Owner	Intact Female	30.7157941956301	-97.4523664870572	104.355654761905
2225	2 years	A757341	Dog	Labrador Retriever Mix	Black/White	2015-09-01	2017-10-03 12:27:00	2017-10-03T12:27:00	19	Partner	Transfer	Intact Female	30.3814182796497	-97.7373217391863	109.074107142857
3319	9 months	A687748	Dog	Labrador Retriever Mix	Yellow	2013-12-09	2014-09-09 17:01:00	2014-09-09T17:01:00		Suffering	Euthanasia	Intact Female	30.729653440489	-97.3912941009824	39.2441468253968
4222	1 year	A735551	Dog	Labrador Retriever Mix	Black	2015-09-25	2016-09-27 14:10:00	2016-09-27T14:10:00	Daisy		Return to Owner	Intact Female	30.5150703656925	-97.6092589677489	52.655753968254

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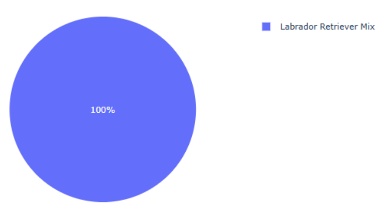
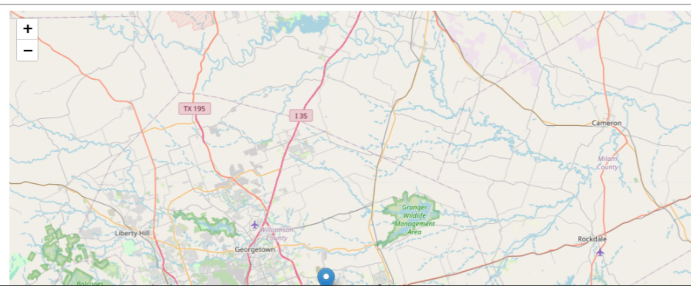
Top 10 Dog Breed Distribution





Mountain or Wilderness Filter

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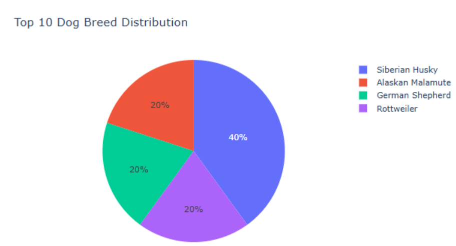
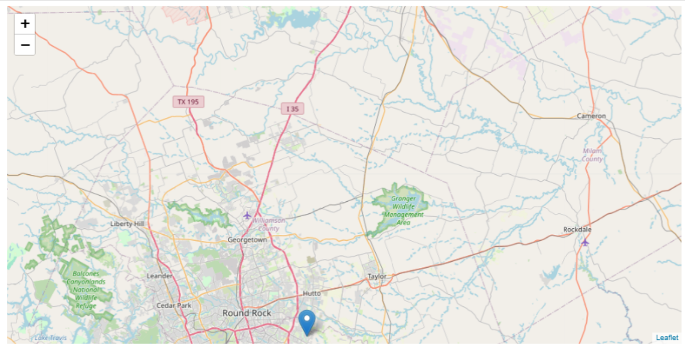
Mountain or Wilderness Rescue

Select a button on the left to update the geolocation map:

	REC_NUM	AGE_UPON_OUTCOME	ANIMAL_ID	ANIMAL_TYPE	BREED	COLOR	DATE_OF_BIRTH	DATETIME	MONTHYEAR	NAME	OUTCOME_SUBTYPE	OUTCOME_TYPE	SEX_UPON_OUTCOME	LOCATION_LAT	LOCATION_LONG	AGE_UPON_OUTCOME_IN_WEEKS
	Search															
○	5315	2 years	A708726	Dog	Alaskan Malamute	Sable/White	2013-07-30	2015-08-02 17:24:00	2015-08-02T17:24:00	Papa		Return to Owner	Intact Male	30.4309339291938	-97.480825836737	104.817857142857
○	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54:00	2018-01-22T11:54:00	Sargent		Return to Owner	Intact Male	30.40668985085	-97.485680334264	26.6422619047619
◉	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23:00	2017-09-23T11:23:00	Zeke		Return to Owner	Intact Male	30.466577208743	-97.5573520930426	120.924900793651
○	3130	2 years	A721834	Dog	Siberian Husky	Brown/White	2014-03-05	2016-03-23 16:23:00	2016-03-23T16:23:00		Suffering	Euthanasia	Intact Male	30.5680998448899	-97.320550480325	107.09751984127
○	6191	2 years	A704101	Dog	Siberian Husky	Black/White	2013-06-01	2015-06-02 16:41:00	2015-06-02T16:41:00	Lobo		Return to Owner	Intact Male	30.4263764229275	-97.4309581796886	104.527876984127





Disaster or Individual Tracking

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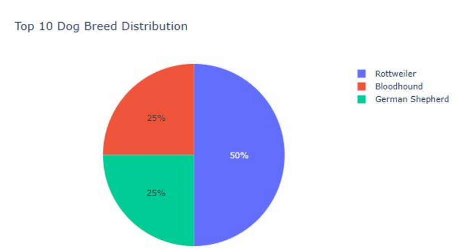
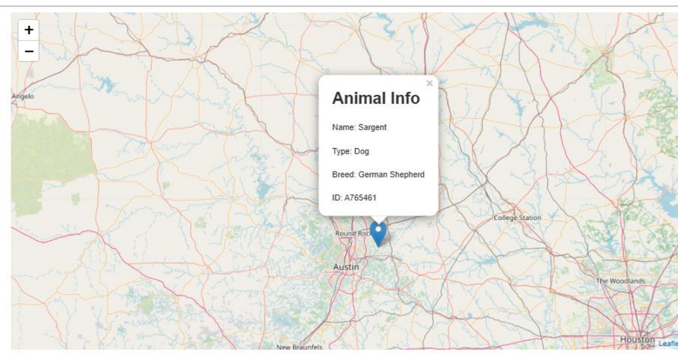
Disaster Rescue or Individual Tracking

Select a button on the left to update the geolocation map:

	REC_NUM	AGE_UPON_OUTCOME	ANIMAL_ID	ANIMAL_TYPE	BREED	COLOR	DATE_OF_BIRTH	DATETIME	MONTHYEAR	NAME	OUTCOME_SUBTYPE	OUTCOME_TYPE	SEX_UPON_OUTCOME	LOCATION_LAT	LOCATION_LONG	AGE_UPON_OUTCOME_IN_WEEKS
	Search															
○	3767	4 years	A712291	Dog	Bloodhound	Red	2011-09-20	2015-09-22 15:43:00	2015-09-22T15:43:00	Boomer		Return to Owner	Intact Male	30.2709983761287	-97.5923916912722	209.093551587302
◉	6557	6 months	A765461	Dog	German Shepherd	Sable	2017-07-20	2018-01-22 11:54:00	2018-01-22T11:54:00	Sargent		Return to Owner	Intact Male	30.40668985085	-97.485680334264	26.6422619047619
○	2987	4 years	A694614	Dog	Rottweiler	Black/Brown	2011-01-01	2015-01-01 14:25:00	2015-01-01T14:25:00	Striker		Return to Owner	Intact Male	30.329873203611	-97.5492968638502	208.800099206349
○	6021	2 years	A728165	Dog	Rottweiler	Black	2015-05-31	2017-09-23 11:23:00	2017-09-23T11:23:00	Zeke		Return to Owner	Intact Male	30.466577208743	-97.5573520930426	120.924900793651





Tools

1. MongoDB -- MongoDB was used for this project to implement the model in the MVC architecture. It accommodates the data needed for the animal shelter, including varied data and location data. By using MongoDB the application will be very easy to use with other datasets due to the flexibility of the document style. MongoDB also offers a lot of support for python, which makes working with the other tools used easier.
2. Python – Various python libraries such as pymongo, dash, pandas and dash leaflet were used in this application due to their flexibility and ease of connecting to MongoDB. In our Python CRUD module, we used pymongo and pandas to connect to Mongo.

    - Pymongo – Used for its ease of use with MongoDB. It offers full support for MongoDB querying. And it is a very scalable library that lets Python dicts be mapped to Mongo's BSON format.
    - Pandas – Pandas is a very powerful data analysis library with the use of data frames. The library helps us connect the data we import from pymongo and display it on dash acting as the controller of the application.
    - Dash – Another powerful python library that acts as the view and controller part of the MVC design. It abstracts most of the front-end work of development allowing us to use built in features to display data such as the data table above.
    - Dash Leaflet – An extension library for dash that lets us implement the geolocation map using the GPS coordinates in the data. This makes the dashboard more interesting and interactive.

Steps to Complete Project

1. Before any work was done on the dashboard, the specs and requirements were heavily reviewed. From the specs and requirements, we created a plan to create the dashboard.
2. The first step of the plan was to use the Python CRUD module to connect with MongoDB and display an unfiltered view of the data. From there, we can add more complex logic.
3. After we added the unfiltered view of the MongoDB data using a dash data table, pymongo, and our CRUD module, we worked on the formatting and styling of the table. We added the ability to perform basic filtering, limited the number of rows, enabled pagination, formatted the headers, added the logo, and so on.
4. Once the data table was displayed neatly and correctly, we created the queries to filter specific types of training dogs. The queries where created from the spec sheet given to us by Grazioso Salvare. Such as water rescue or mountain and wilderness tracking. With these queries, we implemented a dropdown menu so users can easily select what filter

they want to use. With the dropdown and our queries, we can use a dash call back to change the data shown on the data table.

5. Next, we implemented the dash leaflet geolocation chart. Again, using a callback and the GPS coordinates in the data, we can display the location of the animal as well as various info about it using a popup.
6. Lastly, we created a pie chart that will show the top 10 dog breeds in the data table. It will change based on the filter used. If a filter is used that removes the dog breeds or animal type from the data table, then no chart will be shown.

Challenges

Throughout this project, we encountered many challenges and obstacles.

- The formatting of the header and logo proved to be difficult. We used flexbox in CSS to neatly display the two next to each other.
- When creating the queries, it required some trial and error to correctly connect the callback using the filter_type variable. In the end, using an if else branch seemed like an easy way to create it. Future improvements to this are likely possible.
- Like formatting the headers and logo, when working on geolocation chart and pie chart we had to research some CSS to style them well. We also reviewed the dash documentation a lot to create the various styling features for the data table. For example, we wanted to make the drop-down filter menu more intuitive for users. We found that dash lets us add a placeholder value so we can let the user know what it does.
- When creating the pie chart, we had to review the pyplot documentation as the plain pie chart was showing data in the entire data set and un-readable. After creating some subsets in pandas and filtering just the top 10 dog breeds, the chart is significantly more readable.